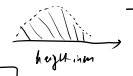
Probability Distribution

height of students = 160, 162, 165, 167, 168, 170, 171, 172, 175

PDF
↳ how densly the probability is distributed



168 - 170 m
↓
higher density in middle

$\boxed{\text{area} = 1}$

$$P(a<X<b) = \int_a^b f(x)\,dx$$

1. Naive Bayes classifier
2. Gaussian distribution

PMF → countable things
CDF → probability accumulated to a point
PDF → measurable thing

You are on AI
↳ transaction
↳ Fraud
↳ Not fraud

Features	Value
Amount	95,000
Time	2:00 AM
Country	USA

$\boxed{\text{PDF}}$

Q1. AI learns from old data
100, 500, 1000, 5000, 10000, 50000, 55000
very few :- 90,000, 95,000, 150000, 200000

AI Qst 95,000
↳ What is the probability density of this value?
PDF(5000) = 0.08
PDF(95000) = 0.0001
↓
unusual → detecting fraud

Distribution
↳ 50 students
↳ 33, 29, 37, 27, 54, 29, 30, 39, 44, 39, 32, 26, 24, 44, 22, 32, 37, 40, 43, 32

40-54 [redacted]
33, 39 [redacted]
22-32 [redacted]

Many many happy returns of the day buddy.

1. Uniform distribution
1 to 6
$\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$
→ office → Singapore
↳ public transport
Bus → 10am to 11am

2. Exponential distribution
Hospitals → ICU ← emergency
P1 P2 P3
mean
10 minutes
5 min
10 min
30 min
lightning

parameter → λ (Lambda)

Normal Distribution
Gaussian distribution

100 students → 145 ⟷ 190 cm
very short people → few
very tall people → few
average people → many
Bell curve

AI → predictes
↳ errors → errors small

Normal distribution
↓
Mean (μ)
↓
e-commerce company
↓
avg. spend → 5000
AI → μ = 5000

standard deviation (σ)
↓

company A	company B
10	5
11	20
10	40
9	10
10	60
small SD	large SD

T shirt
Jean
Shirt
1500 - 3000

Levi's
$\boxed{5000}$

$\boxed{68 - 95 - 99.7}$ rule

R1 68% data lies within one s.d.
μ-σ μ μ+σ
|---- 68% ----|
μ = 170 cm
SD = 10 cm
68% 160 & 180

R2 95% data lies within two s.d.
|------ 95% ------|
μ-2σ μ+2σ
μ = 170 cm
SD = 10 cm
95% 150 & 190 cm

R3 99.7% data lies within three s.d.
|------ 99.7% ------|
μ-3σ μ+3σ
μ = 170 cm
SD = 10 cm
99.7% 140 cm & 200 cm

Standard N.D.

A	B
marks = 80	500
average = 70	450

standardize.

Z - Score :→ how far a value is from average

$$Z = \frac{x-\mu}{\sigma}$$

x → actual value

class average → 70
s.d. (σ) → 10
marks → 80

$$Z = \frac{80-70}{10} = 2$$

student is 2 s.d. above average

avg. body temp. = 98.4
p. tem. = 105

Z score high.
AI alert doctor.

3 → pattern
↳ data → cost

Cybersecurity AI → 5 500 Z score ↑
AI → hacking.

t-test
5 student
↳ little
↳ N.D.
t distribution → N.D.
↓
heavy tails
↓
extreme values

Chi-squared distribution
↓
square → positive

2 = 4
-2 = 4

Student	Marks
A	48
B	50
C	52

average = 50

-2, 0, 2 ⇒ 4, 0, 4

right skewed

F-distribution
A → 50, 51, 49, 50, 50
B → 40, 70, 30, 80, 90

F dist.

$$F = \frac{V_1}{V_2}$$

F is also +ve.

X_1, X_2

$$F = \frac{X_1^2/df_1}{X_2^2/df_2}$$